Edge Inference Solution Package — key decisions (v1.0, 11 Oct 2026)

- **Runtime:** LiteRT int8 (XNNPACK CPU), 7 MB YOLO model — CPU-only 4-core x86, 2 GB RAM. Fallback: ONNX Runtime CPU (fp32/QDQ-int8) if int8 misses mAP@0.5 0.88.
- **Edge runtime:** Azure IoT Edge 1.6 LTS — edgeAgent/edgeHub `1.6.0`, aziot packages held at `1.6.*`, bootstrap agent 1.6.0; 1.5 LTS (EoS 10 Nov 2026) excluded.
- **Host OS:** Ubuntu Server 22.04 (Tier 1). Azure Linux 3.0 is not on the 1.6 supported list → container base image only.
- **Offline buffer:** edgeHub store on dedicated 8 GB volume (5 GB budget + 1 GB spool + 2 GB headroom); routes P0 damage / P1 results TTL 7 d, P2 health TTL 1 h; 72 h = 1.2 GB (24 %); 80 % alarm = abnormal.
- **Sync:** AMQP 1.0/TLS upstream; IoT Hub S2 × 2 (≈5.3 M of 12 M msgs/day); deterministic idempotency key, 14-day dedupe; drain ⩽ 15 min (4 h), ⩽ 60 min (72 h).
- **Deployment:** Canary — Flagger on Gateway API (AKS Istio add-on; Ingress-NGINX avoided) for central; ring rollout R0 1 → R1 5 → R2 20 → R3 50 via layered IoT Hub deployments + KQL gate; base deployment = known-good floor.
- **Recovery:** dashboard actions ($edgeAgent direct methods) + GitHub Revert / Run workflow; kubectl platform-only.
- **Runbook SLAs:** rollout success ⩾ 98 %; MTTR 30 min site / 2 h fleet; escalate at ⩾ 3 concurrent sites.
- **Timeline:** W1–3 build/prove, W4 R0–R1, W5 R2–R3, W6–8 freeze.